nature.com [[light_sheet_microscopy/chhetri_keller_2015/paper|paper]]

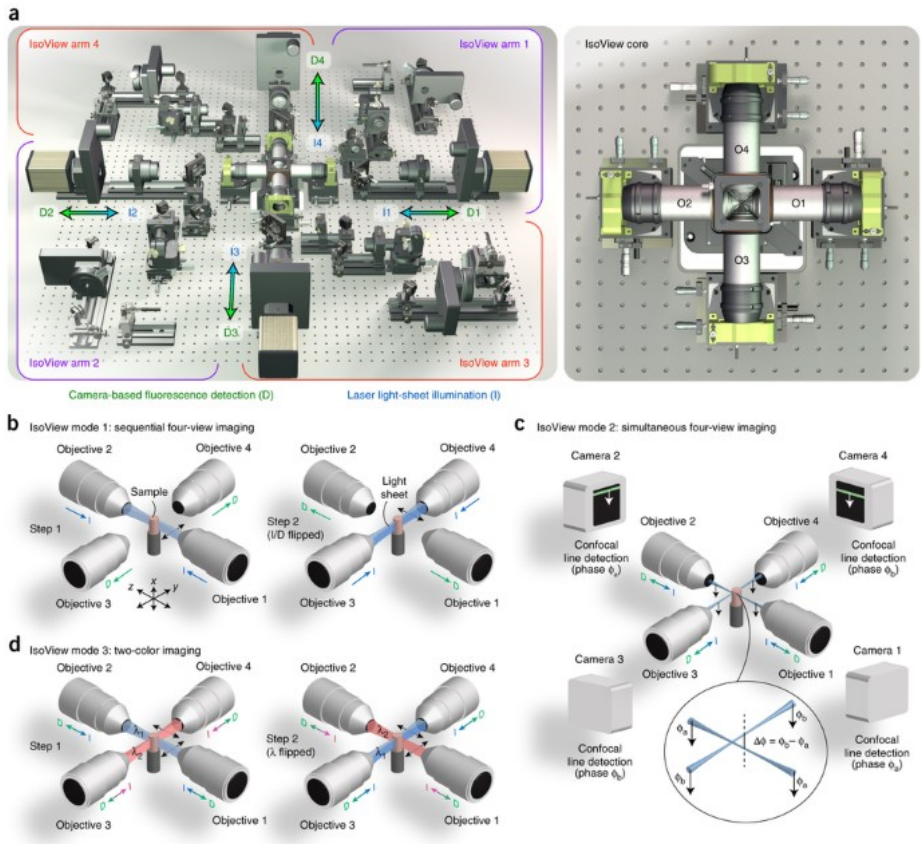
Whole-animal functional and developmental imaging with isotropic spatial resolution

Key takeaways

- Need *at least* 2 additional views (2 -> 4) needed for whole-CNS imaging in larger specimin > 80x80x50um^3
  - 400-fold larger volumes
- Dresophila `@2Hz` with `1.1-2.5um` resolution
- Zebrafish `@1Hz`
- Multicolor gastrulating drosophila `@0.25hz`

The microscope

Figure 1: Isotropic multiview light-sheet microscopy.



- **4 identicle** orthogonally arranged arms - Illuminate with scanned laser light sheet - Image emitted fluoresence

1. PSF before multi-view registration: [[snap1.png]]
2. Inject beads into embryo [[bead_site.png]]
3. Multiview-deconv fixes lateral/axial PSF [[bead_site2.png]]

Location Matters

- signal strengths at locations near the surface of the embryo (< 20 m depth, vs the center ~100 m depth),
- bead fluorescence intensities decrease on average by a factor of 5 in the raw views and by a factor of 6 in the multi-view deconvolved IsoView data
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